

RakshakAI

Predictive Personnel Stress & Welfare Monitoring System

Smart India Hackathon 2026	Problem Statement: SIH26186
Team: ISM Troopers	Team ID: 188472
Category: Software	Theme: HealthTech / BioTech / MedTech

Architecture & Technical Design Document

This document organizes the proposed RakshakAI solution described in the SIH idea submission into a clear system architecture, end-to-end workflow, technology stack, data flow, privacy/security model, and major functional modules.

Source basis: SIH26186 problem statement and the team's Template 2–4 technical approach and feasibility material.

1. System Architecture

RakshakAI is designed as a multi-layer welfare intelligence platform. Operational and wellness inputs are ingested into a Digital Welfare Twin, analyzed by an AI Risk Assessment Engine, and converted into recommendations, alerts, and role-specific dashboards. The architecture preserves privacy by limiting sensitive personal information exposed to command-level views.

INPUT LAYER	INTELLIGENCE LAYER	ACTION & INSIGHT LAYER
HRMS Data Leave • Deployment • Duty • Transfers	Data Ingestion Operational + wellness + optional wearable data	Welfare Intelligence Platform Insights • Recommendations • Intervention
Wellness App Self-assessments • Feedback	Digital Welfare Twin Continuous personnel wellness profile	Welfare Intervention Center Review • Validate • Support • Follow-up
Optional Wearables Sleep • Activity • Health indicators	AI Risk Assessment Engine Explainable AI + rule-based analytics	Commander Readiness Dashboard Unit trends • workload • readiness

Core functional modules

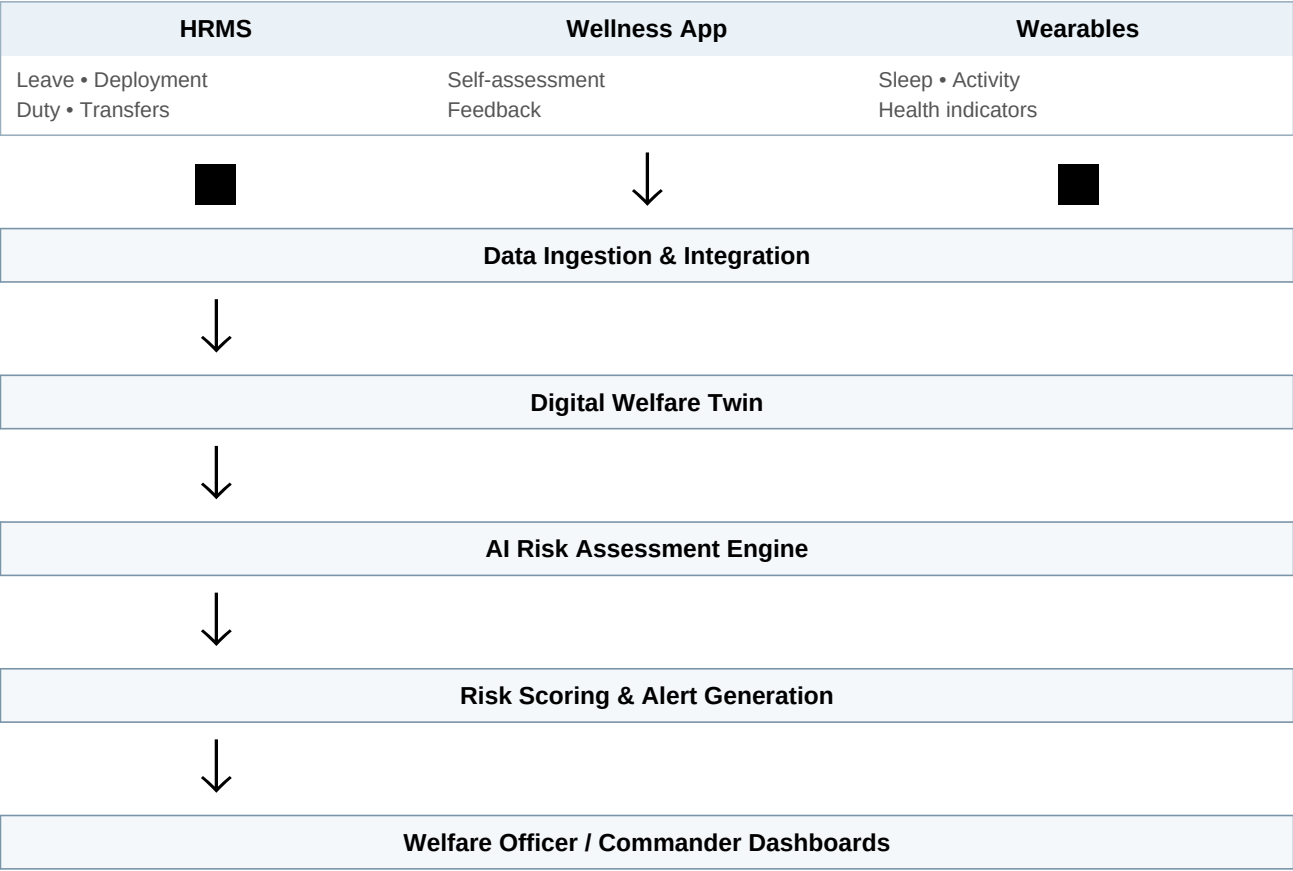
Module	Purpose
Behavioral Risk Analytics	Uses deployment history, leave utilization, duty schedules, transfer frequency, and workload patterns to identify early indicators of stress, fatigue, and burnout.
Personnel Wellness App	Enables self-assessments, wellness trend tracking, AI-generated recommendations, and confidential support requests.
Digital Welfare Twin	Combines operational data, wellness assessments, and behavioral indicators into a dynamic wellness profile for each personnel.
AI Risk Prediction Engine	Applies explainable AI and rule-based analytics to detect stress patterns, burnout risks, fatigue trends, and welfare concerns.
Welfare Intervention Center	Allows welfare officers to review high-risk cases, validate findings, coordinate support, schedule follow-ups, and monitor outcomes.
Commander Readiness Dashboard	Provides unit-level wellness trends, operational readiness insights, workload distribution analytics, and workforce resilience indicators without exposing sensitive personal data.

2. End-to-End Workflow

Step	Flow	Output
1	Collect data from HRMS, Wellness App and optional wearables.	Unified input stream
2	Ingest and normalize operational, wellness and behavioral data.	Analysis-ready data
3	Continuously update the Digital Welfare Twin for each personnel.	Dynamic wellness profile

Step	Flow	Output
4	Run AI models and rule-based analytics for stress, fatigue, burnout and welfare patterns.	Risk indicators
5	Generate risk scores and alerts.	Prioritized cases / alerts
6	Route cases to the appropriate welfare workflow; urgent cases can follow the rapid triage strategy described in the proposal.	Intervention action
7	Provide role-specific insights to welfare officers and commanders.	Support + readiness insights
8	Track intervention outcomes and update the personnel wellness profile.	Continuous monitoring

Simplified data flow



3. Technology Stack

Layer	Technology / Approach	Role in RakshakAI
Frontend	React / React Native	Mobile and web user interfaces for personnel, welfare officers and commanders.
Backend	Node.js / Express / FastAPI	Backend APIs and service layer for application and analytics integration.
Database	MongoDB / PostgreSQL	Persistent storage for operational and application data.
Data Ingestion	HRMS + Wellness App + optional wearables	Collect operational, self-assessment and optional physiological/activity inputs.
ML / Analytics	XGBoost, LSTM, NLP (BERT)	Risk prediction, temporal pattern analysis and language-related analytics.
Risk Layer	Risk scoring + alert generation	Convert model/rule outputs into prioritized welfare alerts.
Dashboard	Welfare / Commander UI	Present intervention workflows, unit-level trends and readiness insights.

4. Privacy & Security Architecture

Control	Design described in the SIH proposal
Role-Based Access Control	Access is separated by role so welfare and command users receive appropriate information.
Consent Management	Explicit consent is part of the welfare-monitoring protocol.
Data Anonymization	Privacy-preserving analytics and anonymization are included to protect sensitive information.
Encryption	Encryption is identified as a core security measure.
Audit Logs	Audit logging is included to support accountability and traceability.
Privacy from Command Records	The proposal describes a cryptographic privacy shield intended to keep health metrics separated from ACRs and disciplinary files.
Welfare Officer Bridge	The welfare officer role is intended to validate severity and advise commanders while keeping raw personal data private.

5. Key Feasibility Strategies

Challenge	Proposed mitigation
Trust deficit / stigma	Cryptographic Privacy Shield and explicit welfare-monitoring consent.
Biometric noise / environmental drift	Cross-analysis of vitals with resilience history, leave deprivation and night-shift frequency.

Challenge	Proposed mitigation
Disconnected hierarchy	Welfare Officer Air-Gapped Bridge between personnel and command-level decision support.
Crisis approval delays	75% Rapid Triage Protocol described in the proposal for high-stress alerts.
Immediate field decompression needs	In-situ resilience tools such as 4-4-4-4 breathing counters and post-patrol relaxation flows.

6. Architecture Summary

RakshakAI follows a closed-loop welfare monitoring model: collect relevant operational and wellness signals, build a continuously updated Digital Welfare Twin, analyze risk using AI and rule-based methods, generate prioritized alerts, route cases for welfare intervention, and use outcomes to maintain the personnel wellness profile. The proposal emphasizes privacy, role separation, explainability, and unit-level readiness insights.

Important: This document reflects the technical approach stated in the supplied SIH material. Where the proposal lists alternatives (for example, Node.js/Express/FastAPI or MongoDB/PostgreSQL), the alternatives are preserved rather than presenting one as already finalized.